

# Nash Equilibrium

Definition, existence, computation, and multiplicity of Nash equilibrium — the central solution concept in non-cooperative game theory.

## Learning objectives

- Define Nash equilibrium in terms of best responses and unilateral deviations.
- State Nash's existence theorem and understand why mixed strategies guarantee existence.
- Compute pure-strategy and mixed-strategy Nash equilibria for 2×2 games in R.
- Visualize best-response correspondences and identify equilibria graphically.

## Motivation

Consider two competing firms choosing prices simultaneously. Each firm wants to maximize profit, but profit depends on the rival's price. Firm A cannot simply optimize in isolation — it must reason about Firm B's choice, knowing that Firm B is reasoning about Firm A in exactly the same way.

This mutual consistency requirement is the heart of Nash equilibrium: a strategy profile where no player can improve their payoff by changing their strategy alone, given what everyone else is doing. It is the most widely used solution concept in non-cooperative game theory, and the starting point for almost every applied analysis of strategic interaction.

## Theory

### Best responses

Given a finite game  $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$  with player set  $N$ , action sets  $A_i$ , and payoff functions  $u_i$ , player  $i$ 's **best response** to a strategy profile  $a_{-i}$  of the other players is any action that maximizes  $i$ 's payoff:

$$BR_i(a_{-i}) = \arg \max_{a_i \in A_i} u_i(a_i, a_{-i}) (\#eq : best - response) \quad (1)$$

### Nash equilibrium defined

#### Definition: Nash Equilibrium

A strategy profile  $a^* = (a_1^*, \dots, a_n^*)$  is a **Nash equilibrium** if every player's strategy is a best response to the others:

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*) \quad \forall a_i \in A_i, \forall i \in N (\#eq : nash - def) \quad (2)$$

Equivalently,  $a_i^* \in BR_i(a_{-i}^*)$  for all  $i$ .

The definition captures a stability property: at a Nash equilibrium, no player has a profitable **unilateral** deviation. This does not mean the outcome is socially optimal (the Prisoner's Dilemma shows otherwise), nor that players coordinate easily — it means the profile is self-enforcing once reached.

## Existence

Nash [Nash1950] proved that every finite game — one with finitely many players and finitely many actions per player — has at least one Nash equilibrium, possibly in mixed strategies. The proof uses Kakutani's fixed-point theorem applied to the best-response correspondence.

### Theorem: Nash's Existence Theorem

Every finite game has at least one Nash equilibrium in mixed strategies.

The theorem does not guarantee a *pure-strategy* equilibrium. The game Matching Pennies, for instance, has no pure-strategy equilibrium but has a unique mixed equilibrium where each player randomizes 50-50 (see [sec-mixed-strategies]).

## Multiplicity

Games can have zero, one, or many pure-strategy equilibria:

- **Matching Pennies** — zero pure, one mixed.
- **Battle of the Sexes** — two pure, one mixed.
- **Prisoner's Dilemma** — one pure (Defect, Defect), which is also the unique NE overall.

When multiple equilibria exist, selecting among them is a separate problem — the **equilibrium selection** problem — addressed by refinements such as subgame perfection ([sec-extensive-form]), trembling-hand perfection, and risk dominance.

## Implementation in R

We implement two approaches: a pure-R solver for 2×2 games and a visualization of best-response correspondences.

### Finding pure-strategy Nash equilibria

```
source(here::here("R", "solvers.R"))

# Prisoner's Dilemma payoff matrices
# Rows = Player 1 actions (C, D), Cols = Player 2 actions (C, D)
A <- matrix(c(3, 0, 5, 1), nrow = 2, byrow = TRUE,
            dimnames = list(c("C", "D"), c("C", "D")))
B <- matrix(c(3, 5, 0, 1), nrow = 2, byrow = TRUE,
            dimnames = list(c("C", "D"), c("C", "D")))

cat("Player 1 payoffs:\n")
```

```
#> Player 1 payoffs:
```

```
A
```

```
#>   C D
#> C 3 0
#> D 5 1
```

```
cat("\nPlayer 2 payoffs:\n")
```

```
#>
```

```
#> Player 2 payoffs:
```

```
B
```

```
#>   C D
```

```
#> C 3 5
```

```
#> D 0 1
```

```
pure_ne <- solve_2x2_pure_nash(A, B)
```

```
cat("\nPure-strategy Nash equilibria:\n")
```

```
#>
```

```
#> Pure-strategy Nash equilibria:
```

```
for (eq in pure_ne) {  
  cat(sprintf(" (%s, %s) with payoffs (%d, %d)\n",  
              rownames(A)[eq[1]], colnames(A)[eq[2]],  
              A[eq[1], eq[2]], B[eq[1], eq[2]]))  
}
```

```
#>   (D, D) with payoffs (1, 1)
```

## Battle of the Sexes — multiple equilibria

```
# Battle of the Sexes
```

```
A_bos <- matrix(c(3, 0, 0, 2), nrow = 2, byrow = TRUE,  
               dimnames = list(c("Opera", "Football"), c("Opera", "Football")))
```

```
B_bos <- matrix(c(2, 0, 0, 3), nrow = 2, byrow = TRUE,  
               dimnames = list(c("Opera", "Football"), c("Opera", "Football")))
```

```
cat("Battle of the Sexes:\n")
```

```
#> Battle of the Sexes:
```

```
cat("Player 1 payoffs:\n")
```

```
#> Player 1 payoffs:
```

```
A_bos
```

```
#>           Opera Football
```

```
#> Opera         3         0
```

```
#> Football      0         2
```

```
cat("\nPlayer 2 payoffs:\n")
```

```
#>
```

```
#> Player 2 payoffs:
```

```
B_bos
```

```
#>      Opera Football
```

```
#> Opera      2      0
```

```
#> Football  0      3
```

```
pure_bos <- solve_2x2_pure_nash(A_bos, B_bos)
```

```
mixed_bos <- solve_2x2_mixed_nash(A_bos, B_bos)
```

```
cat("\nPure-strategy NE:\n")
```

```
#>
```

```
#> Pure-strategy NE:
```

```
for (eq in pure_bos) {  
  cat(sprintf(" (%s, %s)\n", rownames(A_bos)[eq[1]], colnames(A_bos)[eq[2]]))  
}
```

```
#> (Opera, Opera)
```

```
#> (Football, Football)
```

```
cat(sprintf("\nMixed-strategy NE: p = %.2f, q = %.2f\n",  
  mixed_bos$p, mixed_bos$q))
```

```
#>
```

```
#> Mixed-strategy NE: p = 0.60, q = 0.40
```

## Best-response correspondence plot

```
# For a 2x2 game, let p = P(Player 1 plays row 1), q = P(Player 2 plays col 1)  
# Player 2's expected payoff from col 1: p * B[1,1] + (1-p) * B[2,1]  
# Player 2's expected payoff from col 2: p * B[1,2] + (1-p) * B[2,2]  
# Player 2 is indifferent when these are equal
```

```
p_seq <- seq(0, 1, length.out = 200)
```

```
# Player 1's best response: BR1(q)  
# EU1(row1) = q*A[1,1] + (1-q)*A[1,2] = q*3  
# EU1(row2) = q*A[2,1] + (1-q)*A[2,2] = (1-q)*2  
# Indifferent when 3q = 2(1-q) => q = 2/5  
br1_q <- 2 / (3 + 2) # q at which P1 is indifferent
```

```
# Player 2's best response: BR2(p)
```

```

# EU2(col1) = p*B[1,1] + (1-p)*B[2,1] = 2p
# EU2(col2) = p*B[1,2] + (1-p)*B[2,2] = 3(1-p)
# Indifferent when 2p = 3(1-p) => p = 3/5
br2_p <- 3 / (2 + 3) # p at which P2 is indifferent

# Build best-response data
br_data <- tibble(
  q = c(0, br1_q, br1_q, 1),
  p_br1 = c(0, 0, 0.5, 1, 1),
  type = "Player 1 BR"
)

br2_data <- tibble(
  p = c(0, br2_p, br2_p, 1),
  q_br2 = c(0, 0, 0.5, 1, 1),
  type = "Player 2 BR"
)

# Equilibrium points
eq_points <- tibble(
  p = c(0, br2_p, 1),
  q = c(0, br1_q, 1),
  label = c("(Football, Football)", "Mixed NE", "(Opera, Opera)")
)

p_fig <- ggplot() +
  geom_path(data = br_data, aes(x = q, y = p_br1),
    colour = okabe_ito[1], linewidth = 1.2) +
  geom_path(data = br2_data, aes(x = q_br2, y = p),
    colour = okabe_ito[2], linewidth = 1.2) +
  geom_point(data = eq_points, aes(x = q, y = p),
    size = 3, colour = okabe_ito[6]) +
  geom_text(data = eq_points, aes(x = q, y = p, label = label),
    vjust = -1, size = 3) +
  annotate("text", x = 0.85, y = 0.15, label = "BR (q)",
    colour = okabe_ito[1], fontface = "bold", size = 4) +
  annotate("text", x = 0.15, y = 0.85, label = "BR (p)",
    colour = okabe_ito[2], fontface = "bold", size = 4) +
  scale_x_continuous(name = "q (Player 2: prob. of Opera)",
    limits = c(-0.05, 1.15), breaks = seq(0, 1, 0.2)) +
  scale_y_continuous(name = "p (Player 1: prob. of Opera)",
    limits = c(-0.05, 1.25), breaks = seq(0, 1, 0.2)) +
  theme_publication() +
  labs(title = "Best-Response Correspondences: Battle of the Sexes")

p_fig

```

```

save_pub_fig(p_fig, "best-response-bos")

```

```

#> Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
#> conversion failure on 'BR (q)' in 'mbcsToSbcs': for (U+2081)

```

```

#> Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :

```

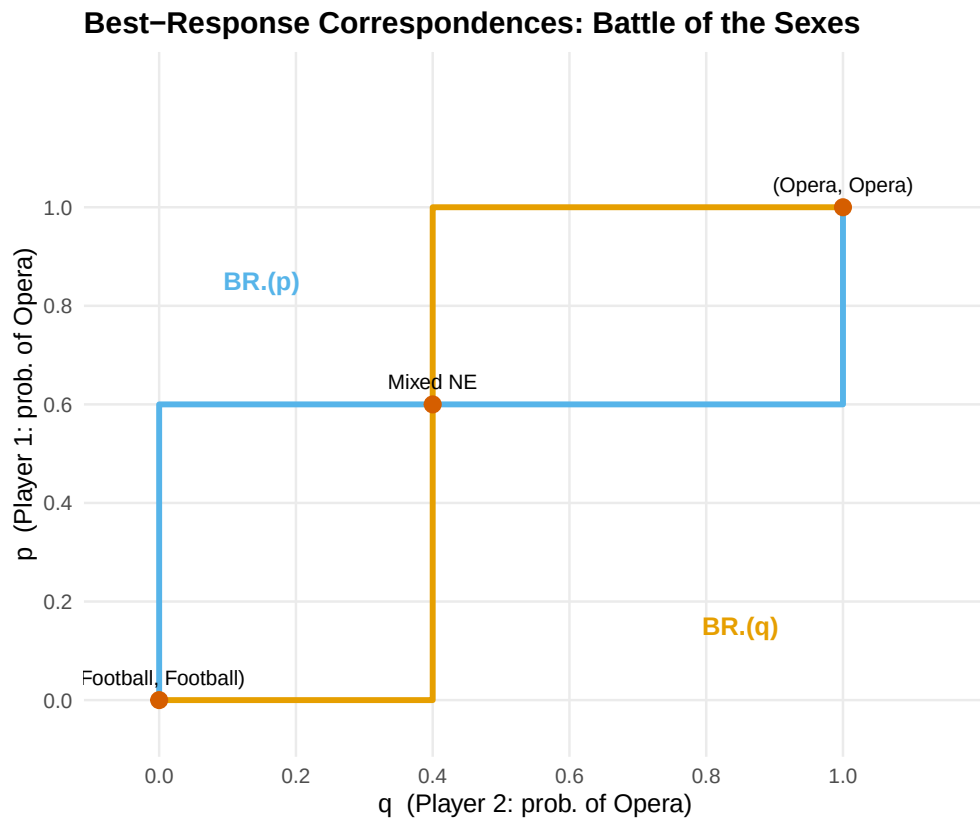


Figure 1: Best-response correspondences for the Battle of the Sexes. Player 1's best response (orange) and Player 2's best response (blue) intersect at three Nash equilibria: two pure-strategy and one mixed.

```
#> conversion failure on 'BR (p)' in 'mbcsToSbcs': for (U+2082)
```

## Worked example

Consider a coordination game where two drivers approach each other on a road and must choose to swerve Left or Right:

```
A_coord <- matrix(c(1, 0, 0, 1), nrow = 2, byrow = TRUE,
                  dimnames = list(c("Left", "Right"), c("Left", "Right")))
B_coord <- A_coord # symmetric game

cat("Coordination Game (symmetric):\n")
```

```
#> Coordination Game (symmetric):
```

```
A_coord
```

```
#>      Left Right
#> Left    1     0
#> Right   0     1
```

```
eq_coord <- support_enumeration(A_coord, B_coord)
```

```
cat("\nPure-strategy NE:\n")
```

```
#>
#> Pure-strategy NE:
```

```
for (eq in eq_coord$pure) {
  cat(sprintf(" (%s, %s)\n", rownames(A_coord)[eq[1]], colnames(A_coord)[eq[2]]))
}
```

```
#> (Left, Left)
#> (Right, Right)
```

```
if (!is.null(eq_coord$mixed)) {
  cat(sprintf("\nMixed NE: p = %.2f, q = %.2f\n",
              eq_coord$mixed$p, eq_coord$mixed$q))
  cat("Expected payoff at mixed NE: ",
      eq_coord$mixed$p * eq_coord$mixed$q * 1 +
      (1 - eq_coord$mixed$p) * (1 - eq_coord$mixed$q) * 1, "\n")
}
```

```
#>
#> Mixed NE: p = 0.50, q = 0.50
#> Expected payoff at mixed NE: 0.5
```

The coordination game has two pure NE — (Left, Left) and (Right, Right) — and a mixed NE at  $p = q = 0.5$  with expected payoff 0.5, which is worse than either pure equilibrium. This illustrates that mixed equilibria can be Pareto-dominated by pure ones: the randomization creates a risk of miscoordination.

## Extensions

Nash equilibrium is the foundation on which most of the rest of this book builds:

- **Mixed strategies** (@ref(sec-mixed-strategies)) formalize the randomization used in the mixed NE above.
- **Extensive-form games** (@ref(sec-extensive-form)) introduce sequential moves and **subgame perfect equilibrium**, a refinement of Nash equilibrium.
- **Bayesian games** (@ref(sec-bayesian-games)) extend Nash equilibrium to settings with incomplete information.
- **Repeated games** (@ref(sec-repeated-games)) show how the folk theorem expands the set of equilibria when games are played repeatedly.

For a comprehensive treatment of Nash equilibrium and its refinements, see @osborne2004 (Chapters 2–4).

## Exercises

1. **Stag Hunt.** Consider the Stag Hunt game with payoffs: both hunt Stag yields  $(4, 4)$ , both hunt Hare yields  $(3, 3)$ , one hunts Stag while the other hunts Hare yields  $(0, 3)$  for the Stag hunter and  $(3, 0)$  for the Hare hunter. Find all pure-strategy and mixed-strategy Nash equilibria. Which equilibrium is risk-dominant? Which is payoff-dominant?
2. **Three equilibria.** Construct a  $2 \times 2$  game that has exactly two pure-strategy Nash equilibria and one mixed-strategy Nash equilibrium. Verify your answer using the `support_enumeration()` function from `R/solvers.R`.
3. **Best-response plot.** Modify the best-response correspondence code in @ref(sec-nash-implementation) to plot the correspondences for the game of Chicken (payoffs:  $(0, 0)$  for mutual swerve,  $(-1, 1)$  and  $(1, -1)$  for one swerving,  $(-5, -5)$  for neither swerving). How many equilibria does the plot reveal?

Solutions appear in @ref(sec-solutions).